Paddy Puddler



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Paddy cultivator & puddler

What it is used for

A paddy puddler mixes soil and water to create soft, muddy field conditions for rice cultivation. It helps prepare a level puddled field before transplanting rice seedlings.

Main parts

- Identify the tractor or power-unit hitching frame.
- Locate the puddling blades, tines or rotary working elements.
- Identify the main drive shaft and transmission components, if fitted.
- Locate the depth or working-position adjustment mechanism.
- Identify the supporting frame and protective guards.
- Locate the hydraulic connections or lifting mechanism, where provided.
- Identify the transport lock and manufacturer's operating instructions.

Before you start

1. Inspect the frame, blades, tines, fasteners and guards for damage.
2. Confirm that the puddler is compatible with the tractor or power unit.
3. Park on firm, level ground, apply the parking brake and stop the engine before attachment.
4. Check hitch pins and secure all connections according to instructions.
5. Inspect the field for stones, stumps, buried objects and people.
6. Ensure water depth and soil conditions are suitable for puddling.
7. Keep bystanders away from the attachment and intended working area.

How to use

1. Position the power unit and puddler on stable ground for attachment.
2. Connect the implement using designated hitch points and secure locking pins.
3. Connect hydraulic or drive components according to the manufacturer's instructions.
4. Set the working depth and implement alignment for the field conditions.
5. Raise the implement fully before travelling to the field.
6. Start the power unit and engage the drive only as instructed.
7. Lower the puddler gradually into the flooded or water-softened soil.
8. Move forward at a suitable low speed while monitoring soil mixing.
9. Raise the implement before turning sharply or crossing obstacles.
10. Stop the power unit, lower the implement safely and inspect after operation.

⚠ Safety

- Keep clear of rotating blades, shafts and the tractor-implement connection.
- Never adjust or clear the puddler while the drive is engaged.
- Stop the engine and lower the implement before inspection or maintenance.
- Never stand beneath a raised implement without approved mechanical support.
- Avoid deep mud, unstable field edges and hidden ditches.
- Use suitable footwear and keep people away from flying mud or debris.

Care & storage

- Remove mud and crop residue after stopping and safely lowering the implement.
- Inspect blades, tines, shafts and fasteners for wear or damage.
- Lubricate designated points and check drive components as instructed.
- Store the implement in a stable, dry location protected from corrosion.

Common mistakes to avoid

- Operating with loose hitch pins or damaged protective guards.
- Working too deeply or too quickly for the soil conditions.
- Turning sharply while the puddler remains engaged in the soil.
- Cleaning mud from moving parts without stopping and isolating the drive.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar